



Sort data with SQL

Hello there! If you're hoping to learn about sorting—in SQL this time— you've definitely come to the right place. So far, we've sorted spreadsheets through the menu and with a written function. Which brings us to the next part of our learning: more sort functions, but this time in SQL. Data analysts love playing with the way data is presented. Sorting is a useful way to rearrange data because it can help you understand the data you have in a different light. As you've probably already noticed, a lot of things you can do in spreadsheets can also be done in SQL.

Sorting is one of those things. We've talked about using SQL with large datasets before. When a spreadsheet has too much data, you can get error messages, or it can cause your program to crash. That's definitely something we want to avoid. SQL shortens processes that would otherwise take a very long time or be impossible to complete in a spreadsheet. Personally, I use SQL to pull and combine different data tables. It's much quicker than a spreadsheet, and that usually comes in handy.

Here's something pretty helpful you can do with SQL. You can use the ORDER BY clause to sort results returned in a query. Let's go back to our movie spreadsheet to get a better idea of how this works. Feel free to follow along in a SQL tool of your choice as we go. As a quick refresher, we have a database of movies listed with data like release

date, director, and more. We can sort this table in lots of different ways using the ORDER BY function. For this example, let's sort by release date.

First, we have the SELECT function and an asterisk. Keep in mind that the asterisk means all columns are selected. Then we have FROM and the name of the database and table we're in right now. Now let's check out the next line. It's empty, but that's where we'll write our ORDER BY function. The ORDER BY command is usually the last clause in your query. Back to the actual sorting!

We'll type ORDER BY with the space. With this clause, you can choose to order data by fields in a certain column. Because we want to sort by release date, we'll type Release_Date. By default, the ORDER BY clause sorts data in ascending order. If you run the query as it is right now, the movies will be sorted from oldest to the most recent release dates. Let's run the query and see what we've got. You can also sort the release dates in the reverse order from the most recent dates to the oldest.

To do this, just specify the descending order in the ORDER BY command written as DESC, D-E-S-C. Let's run this query. As you'll notice, the most recently released films are now at the top of the database. In spreadsheets, you can combine sorts and filters to display information differently. You can do something similar in SQL too. You might remember that while sorting puts data in a specific order, filters narrow down data, so you only see data that fits the filter. For example, let's say we want to filter movies by genre so that we're only working with comedies.

But we still want release dates to be sorted in descending order, from most recent to oldest films. We can do this with the WHERE clause. Let's try that now. First, we'll check that the ORDER BY clause is always the last line. That makes sure that all the results of

the query you're running are sorted by that clause. Then, we'll add a new line for the WHERE clause after FROM and before ORDER BY. Here's what we've got so far.

From there, we want to type the column we're filtering for. In this case, we want to filter the database for comedies. After the WHERE clause, we'll type the column list's name as Genre. Now, we'll add an equal sign after Genre because we only want to include genres that match what we're filtering for. In this case, we're filtering for comedy, so we'll type Comedy between two apostrophes. Now, if you check out the entire query as a whole, you'll notice that we're selecting all columns, and we know it's all columns because that's what an asterisk means. The FROM clause specifies the name of the movie database we're using, and the WHERE clause filters the data to include entries whose genre is specified as comedy.

Then in the last line, we have the ORDER BY clause, which will sort the data we've chosen to filter by release dates in descending order. This means when we run the query, we'll only have comedy movies listed from newest releases to oldest releases. Let's run it and figure out if that's the case. Cool. Check out all those comedy movies and the way those dates are sorted. Now, let's take this query a step further. We'll filter for two conditions at once using the AND filter.

Working off the query we've been using, we'll add a second condition in the WHERE clause. We'll keep the sorting the same. Let's say you wanted to filter by comedy movies and movies that earned over 300 million in the box office. In this case, after the AND function, you'd add the revenue condition by typing Revenue. From there, you'll specify that you only want to return films with revenues over \$300 million. To do that,

type the greater than sign and then the complete number of 300 million without commas. Now let's run the query.

Here, the data only shows comedy movies with revenues of over \$300 million, and it's sorted in descending order by release date. It looks really good. You just filtered and sorted a database like it's your job. And with practice, one day it can be. Just like that, you've finished another step in your data analyst journey. By now, you really **dug** and learned about the analysis process with a special **emphasis** on how organization can change how you go through your data. You also learned about both spreadsheets and SQL, and how to sort and filter data in both types of programs.

To help you get more comfortable using spreadsheet and SQL features, you'll be getting some materials you can use as a resource. Coming up, we'll check out how an organizational mindset can take your analytical skills even further. We'll also cover converting, formatting, and adjusting data to combine information in a way that makes sense. Learning those skills early on can make your work as a data analyst much more efficient and effective in the long run. See you soon.

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